Lecture 9

Naïve Bayes; Review and Final Thoughts

DSC 40A, Summer 2026

Lecture 9 Part 1

Naïve Bayes

DSC 40A, Summer 2026

Agenda

- Classification.
- Classification and conditional independence.
- Naïve Bayes.

Recap: Bayes' Theorem, independence, and conditional independence

- Bayes' Theorem describes how to update the probability of one event, given that another event has occurred.

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- A and B are **independent** if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- A and B are **conditionally independent** given C if:

$$\mathbb{P}((A \cap B)|C) = \mathbb{P}(A|C) \cdot \mathbb{P}(B|C)$$

  - In general, there is no relationship between independence and conditional independence.

Question 🤔

Answer in chat.

Remember, you can always ask questions in Q&A!

Classification

Classification problems

- Like with regression, we're interested in making predictions based on data (called **training data**) for which we know the value of the response variable.
- The difference is that the response variable is now **categorical**.
- Categories are called **classes**.
- Example classification problems:
    - Deciding whether a patient has kidney disease.
    - Identifying handwritten digits.
    - Determining whether an avocado is ripe.
    - Predicting whether credit card activity is fraudulent.
    - Predicting whether you'll be late to school or not.

Example: Avocados

color	ripeness
bright green	unripe ❌
green-black	ripe ✅
purple-black	ripe ✅
green-black	unripe ❌
purple-black	ripe ✅
bright green	unripe ❌
green-black	ripe ✅
purple-black	ripe ✅
green-black	ripe ✅
green-black	unripe ❌
purple-black	ripe ✅

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

Strategy: Calculate two probabilities:

$$\mathbb{P}(\text{ripe} \mid \text{green-black})$$

$$\mathbb{P}(\text{unripe} \mid \text{green-black})$$

Then, predict the class with a **larger** probability.

Estimating probabilities

- We would like to determine $\mathbb{P}(\text{ripe}|\text{green-black})$ and $\mathbb{P}(\text{unripe}|\text{green-black})$ for all avocados in the universe.
- All we have is a single dataset, which is a **sample** of all avocados in the universe.
- We can estimate these probabilities by using sample proportions.

$$\mathbb{P}(\text{ripe}|\text{green-black}) \approx \frac{\#\text{ ripe green-black avocados in sample}}{\#\text{ green-black avocados in sample}}$$

- Per the **law of large numbers** in DSC 10, larger samples lead to more reliable estimates of population parameters.

Bayes' Theorem for Classification

- Suppose that A is the event that an avocado has certain features, and B is the event that an avocado belongs to a certain class. Then, by Bayes' Theorem:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- More generally:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- What's the point?
  - Usually, it's not possible to estimate $\mathbb{P}(\text{class}|\text{features})$ directly.
  - Instead, we often have to estimate $\mathbb{P}(\text{class})$, $\mathbb{P}(\text{features}|\text{class})$, and $\mathbb{P}(\text{features})$ separately.

Example: Avocados

color	ripeness
bright green	unripe ❌
green-black	ripe ✅
purple-black	ripe ✅
green-black	unripe ❌
purple-black	ripe ✅
bright green	unripe ❌
green-black	ripe ✅
purple-black	ripe ✅
green-black	ripe ✅
green-black	unripe ❌
purple-black	ripe ✅

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

Shortcut: Both probabilities have the same denominator, so the larger probability is the one with the **larger numerator**.

$$\mathbb{P}(\text{ripe}|\text{green-black}) =$$

$$\mathbb{P}(\text{unripe}|\text{green-black}) =$$

Classification and conditional independence

Example: Avocados, but with more features

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Strategy: Calculate $\mathbb{P}(\text{ripe} | \text{features})$ and $\mathbb{P}(\text{unripe} | \text{features})$ and choose the class with the **larger** probability.

Issue: We have not seem a firm green-black Zutano avocado before, which means that the following probabilities are undefined:

$$\mathbb{P}(\text{ripe} | \text{firm, green-black, Zutano})$$

$$\mathbb{P}(\text{unripe} | \text{firm, green-black, Zutano})$$

A simplifying assumption

- We want to find $\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$, but there are no firm green-black Zutano avocados in our dataset.
- Bayes' Theorem tells us this probability is equal to:

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano}) = \frac{\mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe})}{\mathbb{P}(\text{firm, green-black, Zutano})}$$

- **Key idea**: **Assume** that features are **conditionally independent** given a class (e.g. ripe).

$$\mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe}) = \mathbb{P}(\text{firm}|\text{ripe}) \cdot \mathbb{P}(\text{green-black}|\text{ripe}) \cdot \mathbb{P}(\text{Zutano}|\text{ripe})$$

Example: Avocados, but with more features

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

$$
\begin{aligned}
\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano}) &= \frac{\mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe})}{\mathbb{P}(\text{firm, green-black, Zutano})} \\
&\propto \mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe}) \\
&= \mathbb{P}(\text{ripe})\mathbb{P}(\text{firm}|\text{ripe})\mathbb{P}(\text{green-black}|\text{ripe})\mathbb{P}(\text{Zutano}|\text{ripe}) \\
&= \left(\frac{7}{11}\right)\left(\frac{1}{7}\right)\left(\frac{3}{7}\right)\left(\frac{2}{7}\right) \\
&= \frac{6}{539}
\end{aligned}
$$

$$\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano}) = \frac{\mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{unripe})}{\mathbb{P}(\text{firm, green-black, Zutano})}$$

$$\propto \mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{unripe})$$

$$= \mathbb{P}(\text{unripe})\mathbb{P}(\text{firm}|\text{unripe})\mathbb{P}(\text{green-black}|\text{unripe})\mathbb{P}(\text{Zutano}|\text{unripe})$$

$$= \left(\frac{4}{11}\right)\left(\frac{3}{4}\right)\left(\frac{2}{4}\right)\left(\frac{2}{4}\right)$$

$$= \frac{3}{44} = \frac{6}{88}$$

Conclusion

- The numerator of $\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$ is $\frac{6}{539}$.
- The numerator of $\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano})$ is $\frac{6}{88}$.
- Both probabilities have the same denominator, $\mathbb{P}(\text{firm, green-black, Zutano})$.
- Since we're just interested in seeing which one is larger, we can ignore the denominator and compare numerators.
- Since the numerator for unripe is **larger** than the numerator for ripe, we **predict that our avocado is unripe** ❌.

Naïve Bayes

The Naïve Bayes classifier

- We want to predict a class, given certain features.
- Using Bayes' Theorem, we write:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- For each class, we compute the numerator using the **naïve assumption of conditional independence of features given the class**.
- We estimate each term in the numerator based on the training data.
- We predict the class with the largest numerator.
    - Works if we have multiple classes, too!

Dictionary

Definitions from Oxford Languages · Learn more

na·ive

/näˈēv/

adjective

(of a person or action) showing a lack of experience, wisdom, or judgment.
"the rather naive young man had been totally misled"

- (of a person) natural and unaffected; innocent.
  "Andy had a sweet, naive look when he smiled"
  Similar: innocent unsophisticated artless ingenuous inexperienced
- of or denoting art produced in a straightforward style that deliberately rejects sophisticated artistic techniques and has a bold directness resembling a child's work, typically in bright colors with little or no perspective.

Example: Avocados, again

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Uh oh!

- There are no soft unripe avocados in the data set.
- The estimate $\mathbb{P}(\text{soft}|\text{unripe}) \approx \frac{\#\text{ soft unripe avocados}}{\#\text{ unripe avocados}}$ is 0.
- The estimated numerator:

  $$\mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{soft, green-black, Hass}|\text{unripe}) = \mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{soft}|\text{unripe}) \cdot \mathbb{P}(\text{green-black}|\text{unripe}) \cdot \mathbb{P}(\text{Hass}|\text{unripe})$$

  is also 0.
- But just because there isn't a soft unripe avocado in the data set, doesn't mean that it's impossible for one to exist!
- **Idea**: Adjust the numerators and denominators of our estimate so that they're never 0.

Smoothing

- **Without** smoothing:

$$\mathbb{P}(\text{soft}|\text{unripe}) \approx \frac{\#\text{ soft unripe}}{\#\text{ soft unripe} + \#\text{ medium unripe} + \#\text{ firm unripe}}$$
$$\mathbb{P}(\text{medium}|\text{unripe}) \approx \frac{\#\text{ medium unripe}}{\#\text{ soft unripe} + \#\text{ medium unripe} + \#\text{ firm unripe}}$$
$$\mathbb{P}(\text{firm}|\text{unripe}) \approx \frac{\#\text{ firm unripe}}{\#\text{ soft unripe} + \#\text{ medium unripe} + \#\text{ firm unripe}}$$

- **With** smoothing:

$$\mathbb{P}(\text{soft}|\text{unripe}) \approx \frac{\#\text{ soft unripe} + 1}{\#\text{ soft unripe} + 1 + \#\text{ medium unripe} + 1 + \#\text{ firm unripe} + 1}$$
$$\mathbb{P}(\text{medium}|\text{unripe}) \approx \frac{\#\text{ medium unripe} + 1}{\#\text{ soft unripe} + 1 + \#\text{ medium unripe} + 1 + \#\text{ firm unripe} + 1}$$
$$\mathbb{P}(\text{firm}|\text{unripe}) \approx \frac{\#\text{ firm unripe} + 1}{\#\text{ soft unripe} + 1 + \#\text{ medium unripe} + 1 + \#\text{ firm unripe} + 1}$$

- When smoothing, we add 1 to the count of every group whenever we're estimating a conditional probability.

Example: Avocados, with smoothing

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Example: Avocados, with smoothing

color	softness	variety	ripeness
bright green	firm	Zutano	unripe
green-black	medium	Hass	ripe
purple-black	firm	Hass	ripe
green-black	medium	Hass	unripe
purple-black	soft	Hass	ripe
bright green	firm	Zutano	unripe
green-black	soft	Zutano	ripe
purple-black	soft	Hass	ripe
green-black	soft	Zutano	ripe
green-black	firm	Hass	unripe
purple-black	medium	Hass	ripe

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

Summary

Summary

- In classification, our goal is to predict a discrete category, called a **class**, given some features.
- The Naïve Bayes classifier uses Bayes' Theorem:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- And works by estimating the numerator of $\mathbb{P}(\text{class}|\text{features})$ for all possible classes.
- It also uses a simplifying assumption, that features are conditionally independent given a class:

$$\mathbb{P}(\text{features}|\text{class}) = \mathbb{P}(\text{feature}_1|\text{class}) \cdot \mathbb{P}(\text{feature}_2|\text{class}) \cdot \ldots$$

Lecture 9 Part 2

More Naïve Bayes, Review

DSC 40A, Summer 2026

Agenda

- Text classification.
  - Practical demo.
- Review.
  - Old exam problems.

Recap: The Naïve Bayes classifier

- We want to predict a class, given certain features.
- Using Bayes' Theorem, we write:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- For each class, we compute the numerator using the **naïve assumption of conditional independence of features given the class**.
- We estimate each term in the numerator based on the training data.
- We predict the class with the largest numerator.
  - Works if we have multiple classes, too!

Text classification

Text classification

- Text classification problems include:
  - Sentiment analysis (e.g. positive and negative customer reviews).
  - Determining genre (news articles, blog posts, etc.).
- Spam filtering is a common text classification problem:

UPR	UltrAsOnic PesT ReSisSer QQ1Q	4:01 AM
HOT97 Summ...	Last Chance To Celebrate 30 Years Of Summer Jam!	6/1/24
Koon Thai kit...	Get 15% OFF At Koon Thai Kitchen	6/1/24
Cori Trattoria...	Thanks San Diego	5/29/24
Smart H0me...	Secure Your Home With Vivint	5/29/24
Smart H0me...	Secure Your Home With Vivint	5/29/24
Pure Barre	What are the Benefits of Pure Barre?	5/29/24
Smart H0me...	Secure Your Home With Vivint	5/29/24

- **Goal**: Given the body of an email, determine whether it's spam or ham (not spam).
- **Question**: What information do we use to make these predictions? What **features**?

Text features

Idea:

- Choose a **dictionary** of d words.
- Represent each email with a **feature vector** $\vec{x}$:

$$\vec{x} = \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \dots \\ x^{(d)} \end{bmatrix}$$

  where $x^{(i)} = 1$ if word i is present in the email, and $x^{(i)} = 0$ otherwise.

This is called the **bag-of-words** model. This model ignores the frequency and meaning of words.

Concrete example

- Dictionary: "prince", "money", "free", and "just".
- Dataset of 5 emails (orange are spam, blue are ham):
  - "I am the prince of UCSD and I demand money."
  - "Tapioca Express: redeem your free Thai Iced Tea!"
  - "DSC 10: free points if you fill out SETs!"
  - "Click here to make a tax-free donation to the IRS."
  - "Free career night at Prince Street Community Center."

	prince	money	free	just	spam?
1					spam
2					ham
3					ham
4					spam
5					ham

Naïve Bayes for spam classification

$$\mathbb{P}(\text{class} \mid \text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features} \mid \text{class})}{\mathbb{P}(\text{features})}$$

- To classify an email, we'll use Bayes' Theorem to calculate the probability of it belonging to each class:
  - $\mathbb{P}(\text{spam} \mid \text{features})$.
  - $\mathbb{P}(\text{ham} \mid \text{features})$.
- We'll predict the class with a larger probability.

Naïve Bayes for spam classification

$$\mathbb{P}(\text{class} \mid \text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features} \mid \text{class})}{\mathbb{P}(\text{features})}$$

- Note that the formulas for $\mathbb{P}(\text{spam} \mid \text{features})$ and $\mathbb{P}(\text{ham} \mid \text{features})$ have the same denominator, $\mathbb{P}(\text{features})$.
- Thus, we can find the larger probability just by comparing numerators:
    - $\mathbb{P}(\text{spam} \mid \text{features}) \propto \mathbb{P}(\text{spam}) \cdot \mathbb{P}(\text{features} \mid \text{spam})$.
    - $\mathbb{P}(\text{ham} \mid \text{features}) \propto \mathbb{P}(\text{ham}) \cdot \mathbb{P}(\text{features} \mid \text{ham})$.

Question 🤔

Answer in chat.

1. $\mathbb{P}(\text{features} \mid \text{spam})$.
2. $\mathbb{P}(\text{features} \mid \text{ham})$.
3. $\mathbb{P}(\text{spam})$.
4. $\mathbb{P}(\text{ham})$.

Which of these probabilities should add to 1?

- A. 1, 2
- B. 3, 4
- C. Both (a) and (b).
- D. Neither (a) nor (b).

Estimating probabilities with training data

- To estimate $\mathbb{P}(\text{spam})$, we compute:

$$\mathbb{P}(\text{spam}) \approx \frac{\#\text{ spam emails in training set}}{\#\text{ emails in training set}}$$

- To estimate $\mathbb{P}(\text{ham})$, we compute:

$$\mathbb{P}(\text{ham}) \approx \frac{\#\text{ ham emails in training set}}{\#\text{ emails in training set}}$$

- What about $\mathbb{P}(\text{features} \mid \text{spam})$ and $\mathbb{P}(\text{features} \mid \text{ham})$?

Assumption of conditional independence

- Note that $\mathbb{P}(\text{features} \mid \text{spam})$ looks like:

$$\mathbb{P}(x^{(1)} = 0, x^{(2)} = 1, \ldots, x^{(d)} = 0 \mid \text{spam})$$

- Recall: the key assumption that the Naïve Bayes classifier makes is that **the features are conditionally independent given the class**.
- This means we can estimate $\mathbb{P}(\text{features} \mid \text{spam})$ as:

$$\begin{aligned} &\mathbb{P}(x^{(1)} = 0, x^{(2)} = 1, \ldots, x^{(d)} = 0 \mid \text{spam}) \\ =&\mathbb{P}(x^{(1)} = 0 \mid \text{spam}) \cdot \mathbb{P}(x^{(2)} = 1 \mid \text{spam}) \cdot \ldots \cdot \mathbb{P}(x^{(d)} = 0 \mid \text{spam}) \end{aligned}$$

Concrete example

- Dictionary: "prince", "money", "free", and "just".
- Dataset of 5 emails (orange are spam, blue are ham):
    - "I am the prince of UCSD and I demand money."
    - "Tapioca Express: redeem your free Thai Iced Tea!"
    - "DSC 10: free points if you fill out SETs!"
    - "Click here to make a tax-free donation to the IRS."
    - "Free career night at Prince Street Community Center."

Concrete example

- New email to classify: "Download a free copy of the Prince of Persia."

	prince	money	free	just	spam?
1	1	1	0	0	spam
2	0	0	1	0	ham
3	0	0	1	0	ham
4	0	0	1	0	spam
5	1	0	1	0	ham

	prince	money	free	just	spam?
1	1	1	0	0	spam
2	0	0	1	0	ham
3	0	0	1	0	ham
4	0	0	1	0	spam
5	1	0	1	0	ham

Uh oh...

- What happens if we try to classify the email "just what's your price, prince"?

	prince	money	free	just	spam?
1	1	1	0	0	spam
2	0	0	1	0	ham
3	0	0	1	0	ham
4	0	0	1	0	spam
5	1	0	1	0	ham

Smoothing

- **Without** smoothing:

$$\mathbb{P}(x^{(i)} = 1 \mid \text{spam}) \approx \frac{\#\text{ spam containing word } i}{\#\text{ spam containing word } i + \#\text{ spam not containing word } i}$$

- **With** smoothing:

$$\mathbb{P}(x^{(i)} = 1 \mid \text{spam}) \approx \frac{(\#\text{ spam containing word } i) + 1}{(\#\text{ spam containing word } i) + 1 + (\#\text{ spam not containing word } i) + 1}$$

- When smoothing, we add 1 to the count of every group whenever we're estimating a conditional probability.

Concrete example with smoothing

- What happens if we try to classify the email "just what's your price, prince"?

	prince	money	free	just	spam?
1	1	1	0	0	spam
2	0	0	1	0	ham
3	0	0	1	0	ham
4	0	0	1	0	spam
5	1	0	1	0	ham

	prince	money	free	just	spam?
1	1	1	0	0	spam
2	0	0	1	0	ham
3	0	0	1	0	ham
4	0	0	1	0	spam
5	1	0	1	0	ham

Modifications and extensions

- **Idea**: Use pairs (or longer sequences) of words rather than individual words as features.
    - This better captures the dependencies between words.
    - It also leads to a much larger space of features, increasing the complexity of the algorithm.
- **Idea**: Instead of recording whether each word appears, record how many times each word appears.
    - This better captures the importance of repeated words.

That's all the new content we have!

Let's now try out Naïve Bayes in code. Follow along here.

Review

We're done with new content! Let's work through some old exam problems.

Fall 2021 Final Exam, Problem 10

Suppose you're given the following probabilities:

- $\mathbb{P}(A|B) = \frac{2}{5}$.
- $\mathbb{P}(B|A) = \frac{1}{4}$.
- $\mathbb{P}(A|C) = \frac{2}{3}$.

Part 1: If A and C are independent, what is $\mathbb{P}(B)$?

Fall 2021 Final Exam, Problem 10

Suppose you're given the following probabilities:

- $\mathbb{P}(A|B) = \frac{2}{5}$.
- $\mathbb{P}(B|A) = \frac{1}{4}$.
- $\mathbb{P}(A|C) = \frac{2}{3}$.

Part 2: Suppose A and C are not independent, and now suppose that $\mathbb{P}(A|\bar{C}) = \frac{1}{5}$. Given that A and B are independent, what is $\mathbb{P}(C)$?

Announcements

Announcements

- Homework 8 is due **tomorrow night**. Solutions will not be released so utilize office hours.
- The Final Exam is on **Friday from 11:30AM-2:30PM PT**.
  - You will be emailed a zoom link soon.
  - 160 minutes, on paper, no calculators or electronics, but **you are allowed to bring one double-sided index card (4 inches by 6 inches) of notes that you write by hand.**
- There is a review session **tomorrow from 6-7:30PM PT in the lecture zoom**.
  - Jordan: confirm poll once it closes and will announce the topics
- If you fill out both the **End-of-Term Survey** and SETs by 8AM on Friday, then you will have 4 points of extra credit added to your HW7.
- Read the proctoring and formatting guidelines posted on Piazza!

Lecture 9 Part 3

Review, Final Thoughts

DSC 40A, Summer 2026

Agenda

- High-level overview of the course.
- Old exam problems.
- Final thoughts.

What was this course about?

Part 1: Empirical risk minimization (Lectures 1-5)

1. Choose a model.

2. Choose a loss function.

3. Minimize average loss to find optimal model parameters.

Part 2: Probability fundamentals (Lecture 6)

- If all outcomes in the **sample space** S are equally likely, then $\mathbb{P}(A) = \frac{|A|}{|S|}$.
- $\bar{A}$ is the **complement** of event A. $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$.
- Two events A, B are **mutually exclusive** if they share no outcomes, i.e. they don't overlap: $\mathbb{P}(A \cap B) = 0$.
- For any two events, the probability that A happens or B happens is $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
- The probability that events A and B both happen is $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A)$.
    - $\mathbb{P}(B|A)$ is the probability that B happens, given that you know A happened.
    - Through re-arranging, we see that $\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}$, which is the definition of conditional probability.

Part 2: Combinatorics (Lectures 6-7)

Suppose we want to select k elements from a group of n possible elements. The following table summarizes whether the problem involves sequences, permutations, or combinations, along with the number of relevant orderings.

	Yes, order matters	No, order doesn't matter
With replacement Repetition allowed	$\boxed{n^k}$ possible **sequences**	more complicated: watch **this video**
Without replacement Repetition not allowed	$\boxed{\frac{n!}{(n-k)!}}$ possible **permutations**	$\boxed{\binom{n}{k}}$ **combinations**

Part 2: The law of total probability and Bayes' Theorem (Lectures 7-8)

- A set of events $E_1, E_2, \ldots, E_k$ is a **partition** of S if each outcome in S is in exactly one E_i.
- The **law of total probability** states that if A is an event and $E_1, E_2, \ldots, E_k$ is a partition of S, then:

$$\mathbb{P}(A) = \mathbb{P}(E_1)\mathbb{P}(A|E_1) + \mathbb{P}(E_2)\mathbb{P}(A|E_2) + \ldots + \mathbb{P}(E_k)\mathbb{P}(A|E_k) = \sum_{i=1}^{k} \mathbb{P}(E_i)\mathbb{P}(A|E_i)$$

- **Bayes' Theorem** states that:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B)\mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- We often re-write the denominator $\mathbb{P}(A)$ in Bayes Theorem' using the law of total probability.

Part 2: Independence and conditional independence (Lecture 8)

- Two events A and B are **independent** when knowledge of one event does not change the probability of the other event.
    - Equivalent conditions: $\mathbb{P}(B|A) = \mathbb{P}(B)$, $\mathbb{P}(A|B) = \mathbb{P}(A)$, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
- Two events A and B are **conditionally independent** given event C if they are independent given the knowledge that event C happened.
    - Condition:
$$\mathbb{P}((A \cap B)|C) = \mathbb{P}(A|C)\mathbb{P}(B|C)$$
- In general, there is no relationship between independence and conditional independence.
- Make sure you've read **this**!

Part 2: Naïve Bayes (Lectures 8-9 (today))

- In classification, our goal is to predict a discrete category, called a **class**, given some features.
- The **Naïve Bayes** classifier works by estimating the numerator of $\mathbb{P}(\text{class} | \text{features})$ for all possible classes.
- It uses Bayes' Theorem:

$$\mathbb{P}(\text{class} | \text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features} | \text{class})}{\mathbb{P}(\text{features})}$$

- It also uses a "naïve" simplifying assumption, that **features are conditionally independent given a class**:

$$\mathbb{P}(\text{features} | \text{class}) = \mathbb{P}(\text{feature}_1 | \text{class}) \cdot \mathbb{P}(\text{feature}_2 | \text{class}) \cdot \ldots$$

Practice problems

Spring 2023 Midterm Exam 2, Problem 6.2

The events A and B are mutually exclusive, or disjoint. More generally, for **any** two disjoint events A and B, show how to express $\mathbb{P}(\bar{A}|(A \cup B))$ in terms of $\mathbb{P}(A)$ and $\mathbb{P}(B)$ **only**.

Fall 2021 Final Exam, Problem 8

Billy brings you back to Dirty Birds, the restaurant where he is a waiter. He tells you that Dirty Birds has 30 different flavors of chicken wings, 18 of which are 'wet' (e.g. honey garlic) and 12 of which are 'dry' (e.g. lemon pepper).

Each time you place an order at Dirty Birds, you get to pick 4 different flavors. The order in which you pick your flavors does not matter.

Part 1: How many ways can we select 4 flavors in total?

Part 2: How many ways can we select 4 flavors in total such that we select an equal number of wet and dry flavors?

Part 3: Billy tells you he'll surprise you with 4 different flavors, randomly selected from the 30 flavors available. What's the probability that he brings you at least one wet flavor and and least one dry flavor?

Part 4: Suppose you go to Dirty Birds once a day for 7 straight days. Each time you go there, Billy brings you 4 different flavors, randomly selected from the 30 flavors available. What’s the probability that on at least one of the 7 days, he brings you all wet flavors or all dry flavors? (Note: All 4 flavors for a particular day must be different, but it is possible to get the same flavor on multiple days.)

Fall 2021 Final Exam, Problem 9

In this question, we'll consider the phone number 6789998212 (mentioned in Soulja Boy's 2008 classic, "Kiss Me thru the Phone").

Part 1: How many permutations of 6789998212 are there?

Part 2: How many permutations of 6789998212 have all three 9s next to each other?

Part 3: How many permutations of 6789998212 end with a 1 and start with a 6?

Part 4: How many different 3 digit numbers with unique digits can we create by selecting digits from 6789998212?

Final thoughts

Learning objectives

On the first day of the quarter, we told you that after taking DSC 40A, you would:

- understand the basic principles underlying almost every machine learning and data science method.
- be better prepared for the math in upper division: calculus, linear algebra, and probability.

What's next?

In DSC 40A, we just scratched the surface of the theory behind data science. In future courses, you'll build upon your knowledge from DSC 40A, and will learn:

- More **supervised learning**, e.g. logistic regression, decision trees, neural networks.
- **Unsupervised learning**, e.g. clustering, PCA.
- More **probability**, e.g. random variables, distributions, stochastic processes.
- More **connections** between all of these areas, e.g. the relationship between probability and linear regression.
- More practical tools.

Thank you!

This course would not have been possible without our TA:

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Congrats on (almost) finishing DSC 40A!

Good luck on the final, and **please keep in touch!**

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